

The Full Monty: Bodies Exposed

Multimodal sensor integration and termination criteria in Monty: an external evaluation

| "Complexity is a symptom of confusion, not a cause." — Jeff Hawkins, *On Intelligence*

Bryan Leavelle 2026-09-12 · <https://github.com/LMSBAND/MontySAI> · MIT license

Abstract

We report an evaluation of the Monty reference implementation [4, 5] conducted entirely through its public extension interfaces; no upstream source was modified. Three sensor modules were added: a cochlear model (CARFAC [1] with a stabilized auditory image [2]), a log-polar retinal model [3], and a depth-only tactile model. Each encodes percept location in a reference frame chosen so that a class of physical transformations becomes a translation, allowing the unmodified evidence-matching learning module to treat those transformations as pose. In a controlled comparison against the reference visual percept recipe (metric 3D locations with HSV and curvature features) under identical patch input, learning-module configuration, and fixation policy, the log-polar representation identified 12 of 12 transformed presentations (scale 1.5×, in-plane rotation 30°, their composition, and out-of-plane tilt to 60°) with no false identifications; the metric representation identified 5 of 12, also without false identifications. Adversarial probes with unlearned objects then exposed three failure modes in the system-level verdict machinery rather than in any sensor: the stock termination criterion counts confident learning modules without requiring agreement on identity; episode termination at the first confident match forecloses evidence that accumulates more slowly; and a functioning sensor that finds no trace of the candidate object has no mechanism for contributing negative evidence. Two modifications, implemented as a subclass and enabled by configuration, reduced false identifications on the probe set from three to zero while preserving recognition of learned objects. A fourth, echolocating sensor extends the method to active sensing: recorded in the frame of the agent's entry pose, a room becomes an object whose recognized pose is the agent's own location within it, and a moving head recovers, from the ears alone, an elevation that a stationary head cannot resolve. We discuss the implications for termination, cross-modal voting, and the representation of absence.

Scope. This document covers the work of 2026-09-12: the retinal and tactile sensors, the representation comparison, the termination analysis, and the echolocation experiments of Section 7. The auditory results it builds on (transposition invariance, cochlear lesion studies, evaluations on recorded guitar) and the acoustic ranging measurements in a physical room were carried out from 2026-09-10 onward and are documented elsewhere in this repository. All claims below correspond to commits, figures, and control conditions available here.

1. Integration method

All components attach to existing extension points. The findings in Section 4 therefore concern default behaviors of the released system, not limitations of its extensibility; the architecture permitted every modification described below.

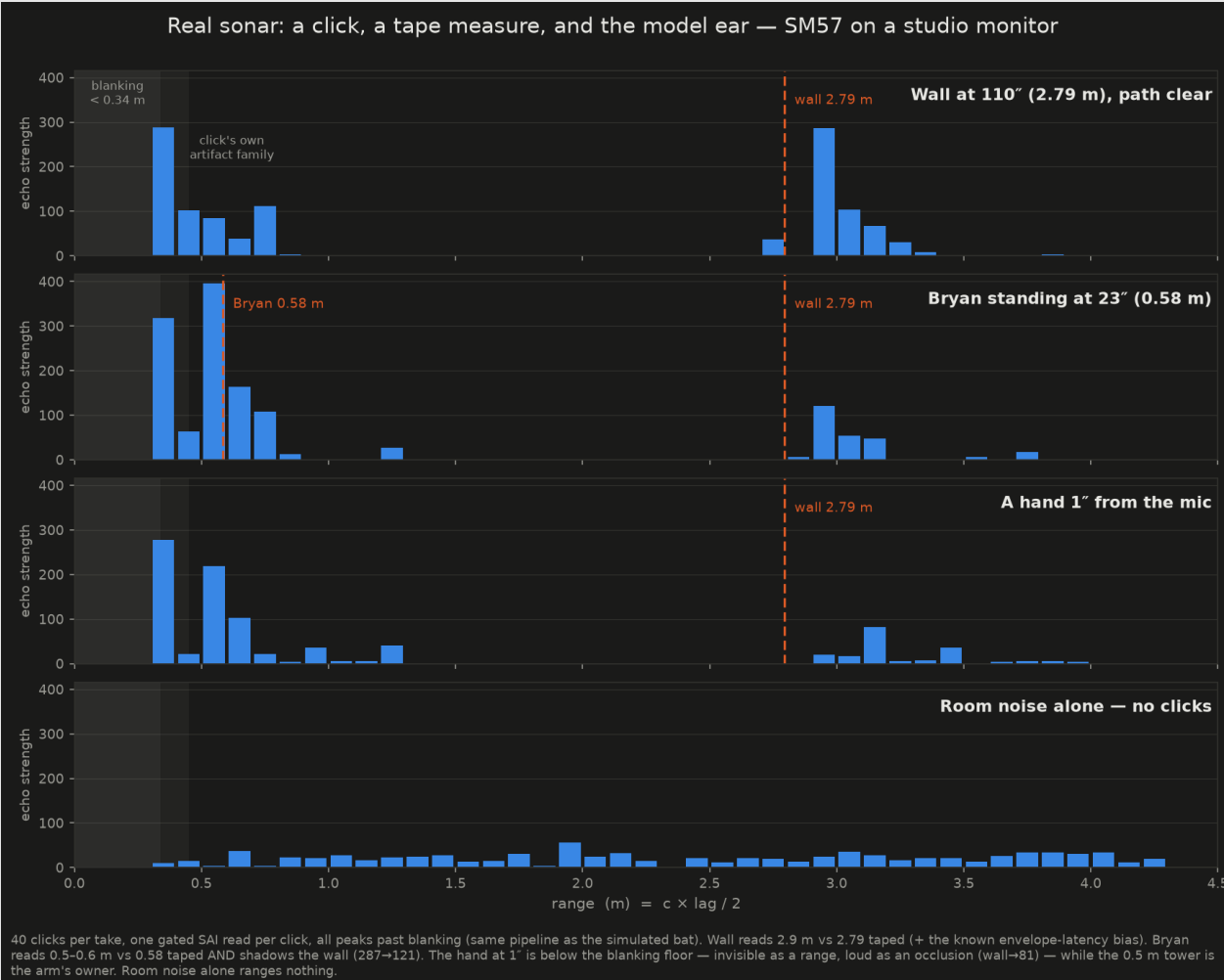
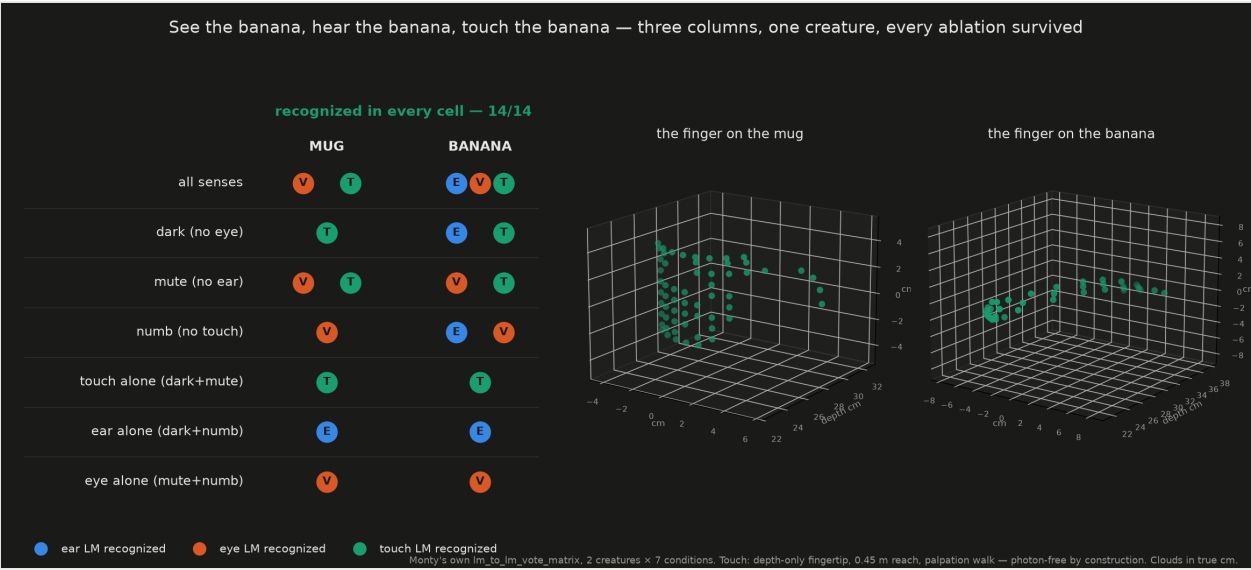
Extension point	Component
SensorModule contract (<code>step</code> , <code>update_state</code> , <code>reset</code> , <code>state_dict</code>)	AudioSM (CARFAC + stabilized auditory image), RetinaSM (log-polar), TouchSM (depth-only), CamSM (reference-recipe control)
Environment composition	AudioEnvironment, which attaches acoustic sources to named objects and renders at the agent pose; the wrapped simulator is unaware of the addition
<code>monty_class</code> configuration key	MontyConsensus, a subclass overriding <code>check_terminal_conditions</code> only
<code>match_criterion</code> configuration	the consensus size N reuses the existing <code>count</code> parameter
<code>lm_to_lm_vote_matrix</code>	three learning modules with all-to-all voting, executed by the unmodified <code>_combine_votes</code>
Hydra configuration tree	all experimental conditions (illumination, audibility, tactile availability, scale, rotation, tilt, probe objects) are configuration changes; no condition is implemented in code

Each sensor follows the same design procedure: select a location frame in which an anticipated transformation acts as a translation; restrict features to a small number of physically interpretable quantities; validate the sensor against the raw simulator before any learning-module involvement; state every recognition claim as a learning-module verdict; and accompany each claim with an ablation matrix.

The three frames:

- **Auditory.** Location is (log-lag in octaves, cochlear place). Pitch transposition is a translation along the log-lag axis, and the learning module recognizes transposed material as the same object at a different pose. This was previously verified on synthetic tones and on recorded guitar performances.
- **Retinal.** Location is (log eccentricity in octaves, polar angle) about a fixation point. Image scaling and image-plane rotation are translations along the two axes. The environment is static, so spatial extent is produced by an internal fixation sequence: at each step the sensor attends to the most salient unvisited cell, with inhibition of return reset per episode.
- **Tactile.** Location is the contact point in meters, recovered by a pinhole model from the depth map. The sensor reads depth only, so it is independent of illumination by construction, and it reports nothing beyond a reach limit of 0.45 m. The reach limit reflects a measured property of the acoustic channel: in recordings made in a physical room (a dynamic microphone and a click track, with distances taped), ranging fails inside approximately 0.5 m because the emission masks its own near field. The tactile modality covers the region where sonar is self-masking.

With three learning modules voting, the system identified both learned objects under seven sensory conditions (all senses; each sense disabled in turn; each sense operating alone), fourteen of fourteen episodes, including tactile-only identification in darkness and silence (Figure 1). The acoustic ranging measurements motivating the tactile reach limit are shown in Figure 2.



2. Comparison of visual representations

The comparison isolates the location frame as the only independent variable. Both visual sensors receive the same 64×64 patch, use the same fixation sequence, and feed identically configured evidence learning modules.

CamSM implements the reference percept recipe: metric 3D locations from depth, surface normal and curvature from local geometry, HSV color as the feature.

CamSM is a reimplementation rather than the reference class itself because `ObservationProcessor.process` requires the observation shape produced by the transform stack (`semantic_3d`, `sensor_frame_data`, `cam_to_world`) and reads the simulator's semantic channel to determine object membership, an oracle unavailable to the other sensors in this study. We have not benchmarked the reimplementation against the original on its own stock configuration; consequently, "the reference recipe" here means that recipe operating under this study's fixation policy and input, not the released class in its native harness.

Configuration parity was audited after an initial run. Pose hypothesis sampling was stock (`initial_possible_poses: "informed"`); `max_nneighbors` was set to 10 where the stock value is 3, a change favoring the reference recipe; the HSV tolerance ([0.1, 0.2, 0.2]) matched the stock benchmark configuration verbatim. The feature weights in the initial run did not (equal weights rather than the stock [2, 0.5, 0.5], which emphasizes hue). Under equal weights the reference recipe produced one false identification (a rotated banana reported as a mug). Under its stock weights that error did not recur, and we withdrew the finding; the final comparison uses stock weights throughout.

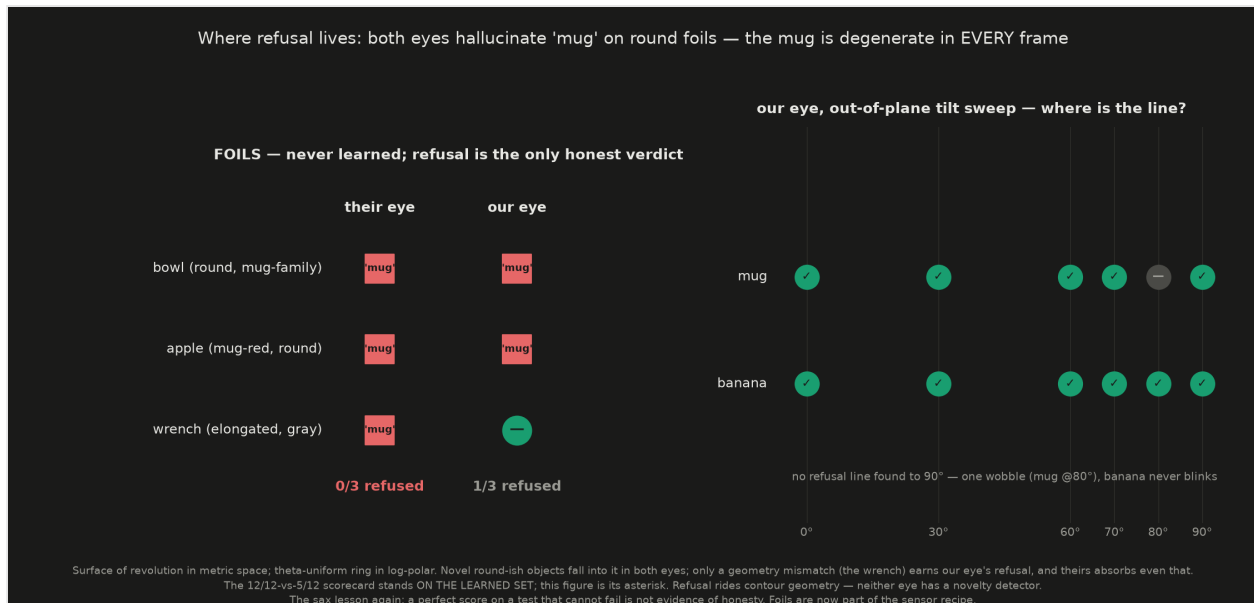
Results over eight in-plane presentations (upright, 1.5× scale, 30° rotation, both) and four out-of-plane presentations (30° and 60° tilt): the log-polar representation identified 12 of 12, with detection at 21 to 31 steps; the metric representation identified 5 of 12. Neither produced a false identification on this learned set. The metric recipe's successes on the scaled and rotated mug are attributable to degeneracy rather than invariance: a surface of revolution with uniform color matches itself under scaling and rotation. The out-of-plane conditions had been registered in advance as the expected advantage of the metric representation, since a log-polar frame has no depth axis; the metric recipe instead declined all four, and the log-polar recipe identified all four (Figure 3). We do not claim this generalizes beyond the two objects tested.

Eye vs eye at THEIR stock tuning — in-plane and out of it. Same patch, same LM machinery, same saccades.			
		their eye 3D metric locations	our eye log-polar locations
upright, 1×	mug	26	21
	banana	21	21
1.5× the size	mug	21	26
	banana	—	21
rotated 30°	mug	26	21
	banana	28	23
1.5× AND 30°	mug	—	21
	banana	—	29
TILT 30° (out-of-plane)	mug	—	21
	banana	—	26
TILT 60° (out-of-plane)	mug	—	31
	banana	—	21
		5/12, zero false	12/12, zero false
recognized (number = steps to verdict) no match (honest refusal) FALSE identification Everything stock on their side: initial_possible_poses='informed', hsv tolerance [0.1,0.2,0.2] and weights [2,0.5,0.5] verbatim from their benchmark yaml, max_neighbors raised 3→10 in their favor. Audit note: at EQUAL feature weights their eye called the rotated banana a mug; their stock 2× hue weight fixes it. The false ID was our tuning, not their eye — retracted. Out-of-plane was pre-registered as the metric eye's home turf; it refused both tilts. Zero false IDs on this learned set — for the foil test both eyes then failed, see where_refusal_lives.png.			

3. Probes with unlearned objects

A recognition score obtained only on learned objects cannot distinguish accuracy from an absence of alternatives. Three objects never presented during training were therefore evaluated: a bowl (a surface of revolution, like the mug), an apple (similar in hue to the mug), and an adjustable wrench (elongated, like the banana). For an unlearned object, declining to identify is the only correct outcome.

Single-sensor results: the log-polar eye identified the bowl and the apple as the mug and declined only the wrench; the metric recipe identified all three as the mug. Both representations therefore admit the same failure class. The mug is the most self-similar learned object in either frame (a surface of revolution in metric space; a ring, uniform in polar angle, in log-polar space), and unlearned objects of approximately round silhouette accrue evidence for it. A tilt sweep to 90° found no comparable failure boundary for learned objects: the log-polar eye continued to identify both learned objects, declining only once (the mug at 80°). Figure 4 summarizes the probe results and the tilt sweep.



4. Findings concerning the verdict machinery

Running the probes through the three-sensor system produced three observations about system-level behavior. Each is reproducible from the configurations in this repository.

4.1. The stock criterion aggregates confidence without requiring agreement on identity. On the bowl, the visual module reached the terminal state "match" with the identity mug while the tactile module reached "match" with the identity banana. `AnyLMsMatch` with `count=2` reports a system-level match under these conditions, since it counts modules in the "match" state without comparing their detected objects. Two modules confident of different identities satisfied the criterion.

4.2. Termination at the first confident match forecloses slower evidence. Verdicts cannot be issued before `min_eval_steps`; in practice the visual and tactile modules are ready at that boundary and the episode ends there. The auditory module identifies the humming mug only when run alone: its percept for a steady tone is nearly a point, all pose hypotheses are equivalent, and the module requires the symmetry-detection path, which needs additional steps of stable evidence. In mixed runs no symmetry event fires before termination. The evidence was available; the episode structure discarded it.

4.3. A functioning sensor cannot contribute negative evidence. On the apple, the visual and tactile modules agreed on the mug, an error of honest degeneracy in both frames. The auditory module was operating throughout and its scene contained no spectral structure at the location the mug hypothesis predicts (a fundamental near 147 Hz). Under the stock protocol this module simply sends no messages, and its lack of corroboration carries no weight. The observation "the predicted signal is not present in an otherwise measured scene" has no representation.

5. Modifications

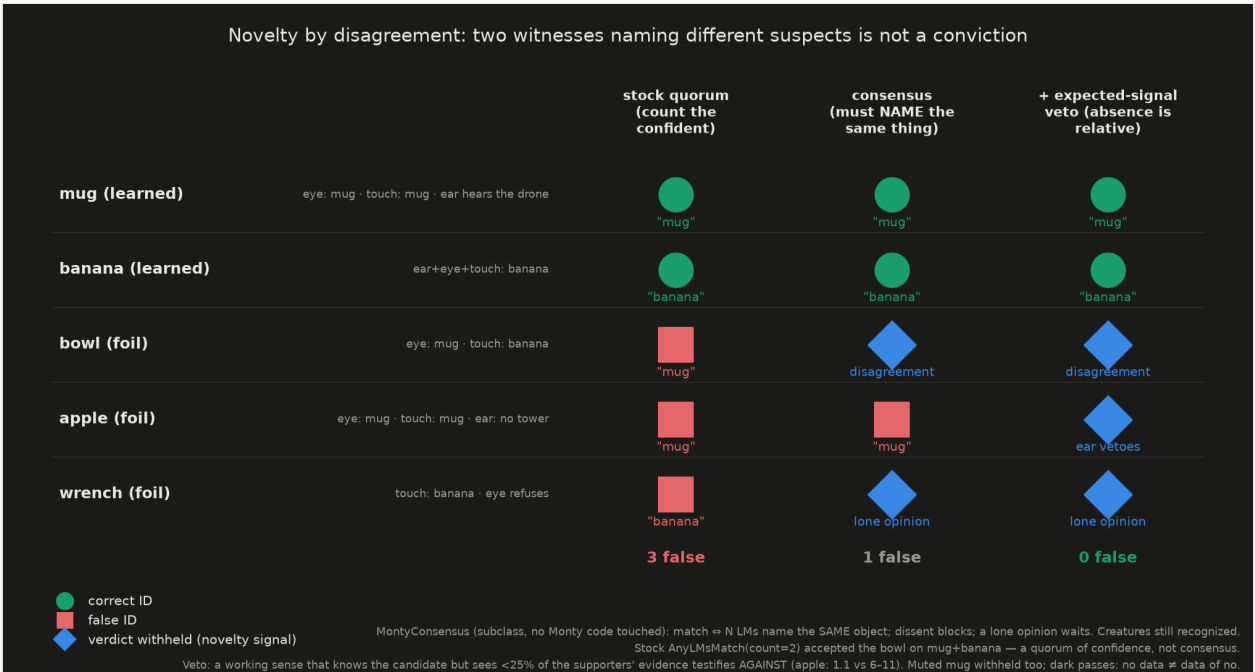
Both changes are implemented in one subclass (`audiomonty/consensus.py`) and enabled by configuration.

Consensus on identity. The system-level match requires at least N modules in the "match" state naming the same object, and it is blocked by any module in the "match" state naming a different object. A single confident

module below N does not terminate the episode. Under this criterion the bowl is blocked by the disagreement between the visual and tactile identifications, and the wrench, supported by one module only, does not terminate. Learned objects remain identified (the mug by two agreeing modules, the banana by three).

A relative test for expected signals. The apple requires a further principle: absence is relative to the scene. An initial implementation tested the dissenting module's evidence against zero and failed, instructively: even in a nominally silent environment the auditory module accrues a small positive evidence value (1.08 in these runs) from the cochlear model's startup transient. The operative test is therefore relative. A module vetoes the verdict when three conditions hold: it is functioning (its sensor is not disabled); it possesses a learned graph for the candidate object; and its evidence for that candidate sits below a fixed fraction (0.25) of the mean evidence of the supporting modules. On the apple, the auditory module held 1.08 against supporting evidence of 6 to 11, and the identification was blocked.

Across the three criteria (stock, consensus, consensus with the expected-signal test) the false identifications on the probe set were 3, 1, and 0 respectively (Figure 5).



The modified criterion has a measurable cost, which we regard as correct behavior rather than a defect. With audio removed from the environment, the auditory module vetoes the identification of the visually and tactually normal mug: the sensor is functioning, the graph predicts a hum, and the scene contains none. Under this protocol an object's acoustic signature is part of its identity. A disabled sensor, by contrast, abstains: darkness does not block identification, because a sensor that receives no input is distinguished from a sensor that measures a scene lacking the predicted structure. The distinction between an absent channel and a measured absence is the substance of both modifications.

The relative threshold also resolves the race of Section 4.2 without further machinery: an auditory module that is genuinely accumulating evidence for the humming mug, but has not yet reached its own terminal state, holds evidence well above the fraction and does not veto.

6. Interpretation

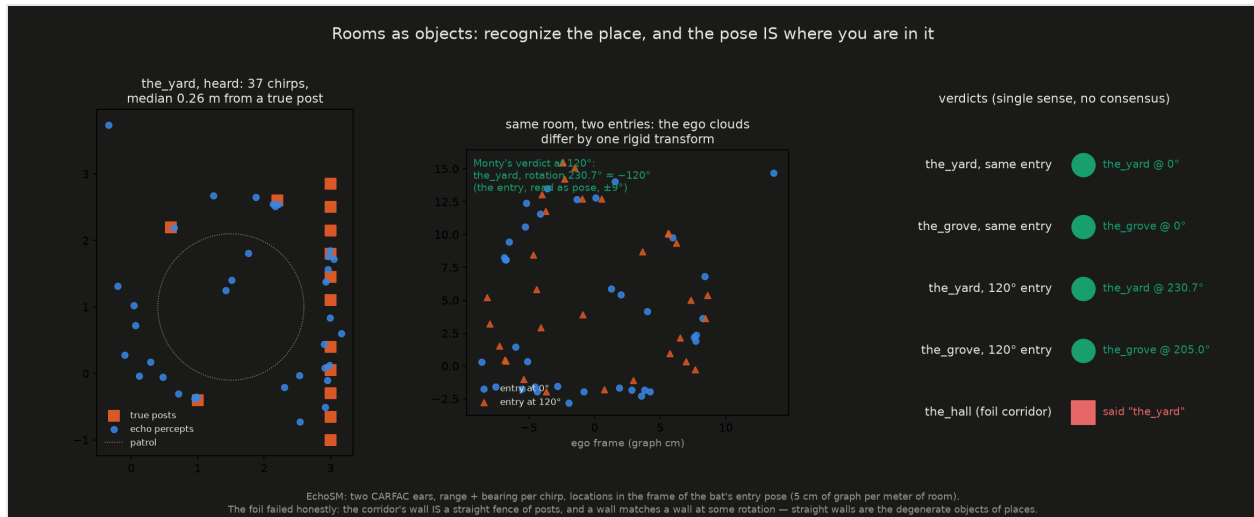
1. **Termination is doing theoretical work.** The failures of Section 4 share a cause: the episode ends at the first confident answer. A recognition process with no terminal state, holding its verdict as a continuously revised quantity, would not exhibit 4.2 at all and would convert 4.1 and 4.3 from structural failures into transient ones. We consider the terminal state an artifact of the evaluation harness rather than a commitment of the underlying theory, and we intend to test this by implementing a recognizer in a context that cannot terminate (a real-time audio process).
2. **The decisive modality is pair-relative.** Each probe was rejected by the channel in which the false hypothesis made its most specific prediction: the apple by audition (the hum), the bowl by tactile shape, the wrench by visual contour. No fixed ranking of modalities describes this; the discriminating channel is the one with the largest divergence between predicted and measured signal for the specific pair of hypotheses in play. Exploiting this requires testing a hypothesis against modalities that did not vote for it, for which the current message flow has no path.
3. **Silence is a measurement.** The startup-transient failure of the absolute threshold is a small demonstration of a general point: a functioning sensor never delivers nothing; it delivers a scene, and the candidate object predicts specific structure standing above that scene's floor. The present environment assigns voices to some objects and leaves the rest acoustically absent, which is physically wrong: in the room recordings noted above, every object measured (a wall, a person, a hand) affected the acoustic field without possessing a voice. Replacing the voiced/voiceless distinction with an ambient acoustic field that all objects scatter is planned; under that model the apple would be discriminated by its scattering signature rather than by veto.
4. **Degeneracy is frame-relative, and every frame has degenerate objects.** The metric frame's mug and the log-polar frame's mug are the same object at its most self-similar in two coordinate systems. Unlearned objects fall toward the most degenerate learned object of any single frame. This locates the remedy at the level of cross-modal aggregation rather than in any improved single sensor.

7. Echolocation, and the case for moving the sensor

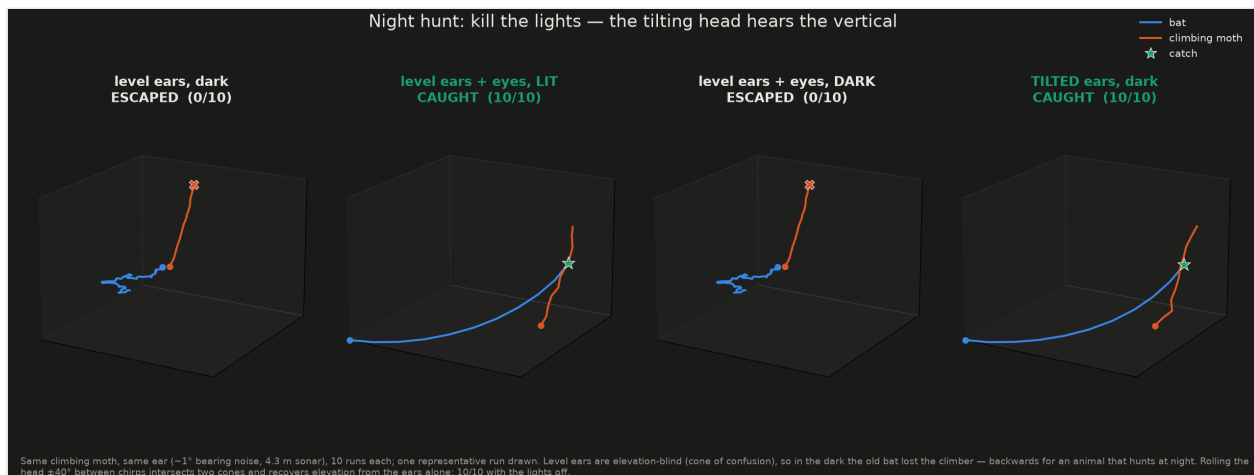
A fourth sensor was built to a different brief. The three above recover their spatial extent from a static scene, either from the evolution of a sound or from an internal fixation sequence; none of them moves. This one does, and moving it turns out to be the entire point. It emits a chirp and listens with two independent CARFAC ears; ranges are read from the spectral-image peaks past an emission-blanking interval, and bearing follows from the interaural range difference (calibrated to about one degree over $\pm 75^\circ$). It was validated twice, in simulation (worst-case ranging error 3.3 cm over 0.5 to 4 m) and physically against the taped distances of Figure 2.

7.1. The room is the object; the pose is where you stand in it. An environment was constructed in which the learned objects are not things but places: a room layout, a set of reflecting surfaces, that a two-eared agent flies a patrol through while chirping. Each chirp yields a range and a bearing, and the reflection is recorded at its position in the frame of the agent's entry pose. This choice of frame is the experiment. In a world frame a room would present an identical map from any point of entry, and recognition would be trivial; in the entry frame, the same room entered from a different point presents a rigidly rotated and translated map, which is exactly the transformation the learning module's pose search exists to undo. Two rooms were learned by flying them. Both were recognized on a fresh flight; and when the agent entered a learned room at a different point of the patrol,

the room was still recognized, and the pose the learning module reported was the agent's actual point of entry, recovered to within about nine degrees (Figure 6). Recognition and self-localization are here a single act: to know the room is to know where in it you stand. The one probe room, a corridor never flown, was misidentified as a learned room at some rotation, for the reason established in Section 6.4: a straight wall matches a straight wall, and straight walls are the degenerate objects of places.

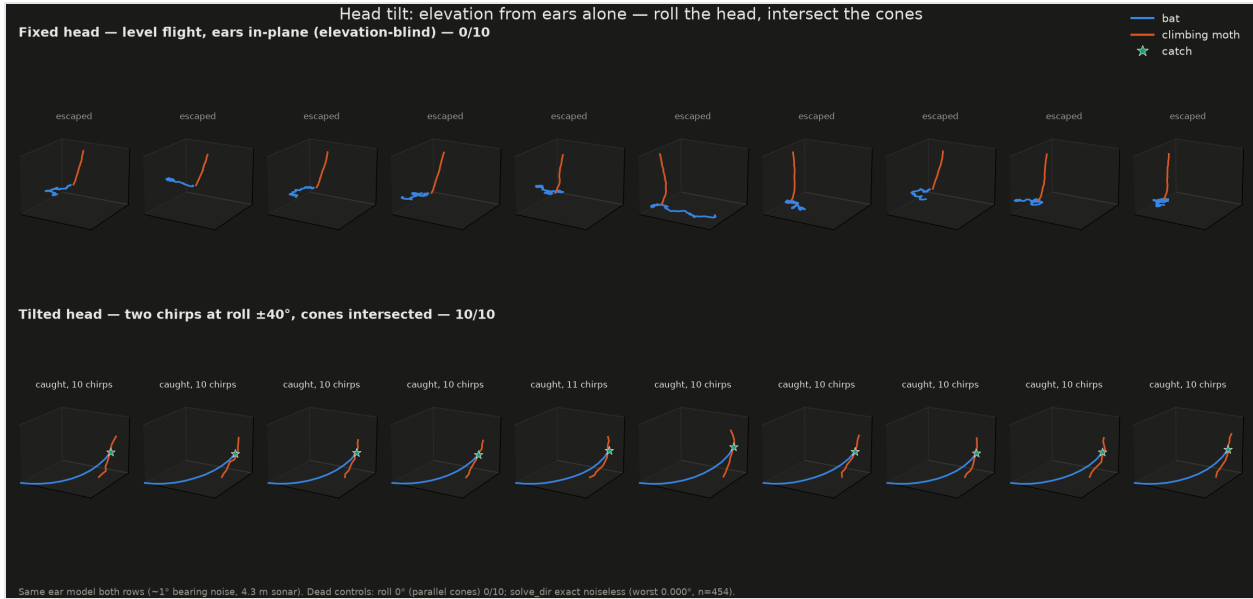


7.2. A level head cannot hear elevation. Two ears on a horizontal baseline measure the difference in path length to a target, which fixes it to a cone about the baseline axis; every direction on that cone, at every elevation, gives the same interaural difference. Against a target that climbs out of the horizontal plane, an agent whose head does not move is blind in precisely the dimension the target is escaping into. Simulated as a pursuit the disadvantage is total: a climbing target is lost on every trial, the pursuer holding the correct azimuth while the target leaves along the one axis the ears cannot resolve (Figure 7, left).



7.3. Rolling the head between chirps recovers it. The cone of confusion is fixed to the head, not to the world. Roll the head about the direction of travel and the baseline axis rolls with it, so a second chirp at a second roll angle fixes the target to a second cone; two cones intersect, generically, in one forward direction, and the elevation the level head could not measure is recovered from the ears alone, with no appeal to vision. The same climbing target that escaped without fail is caught on every trial, in roughly ten chirps (Figure 8). The movement that supplies the elevation is the same movement that would aim a visual field and a directional gain at the target: pointing the head is one motor act serving every sensor on the skull, and the recognition it enables is unavailable to any fixed sensor of equal acuity. This is the sensorimotor thesis of the underlying architecture

at its smallest scale. The movement is the only way to obtain evidence of a kind a stationary sensor cannot reach, in a dimension no amount of listening from one posture would ever supply.



8. Three remarks outside the engineering

8.1. Competence without comprehension. Dennett's phrase [7] describes this work exactly, and we mean that as a finding rather than a disclaimer. The systems above recognize objects, localize themselves, refuse impostors, and register dissent, and at no point does anything in them understand anything. The learning module that recovered a 30° rotation it was never told about contains no concept of rotation; it is displacement bookkeeping over a graph. The veto that behaves like skepticism is a threshold on a ratio of two accumulators. The consensus rule that behaves like judicial caution is a counter and a string comparison. Dennett's claim is that this ordering is the actual architecture of mind: comprehension, where it exists, is composed of competences, and never the reverse. The thousand-brains program is one of the few research efforts that has made this bet explicitly, at the level of architecture, with the cortical column as the unit of competence. Our negative results sharpen the bet in a way we did not anticipate. Every failure documented in Section 4 occurred at a point where the harness had implicitly assumed a comprehension it had never built: the stock criterion behaves as if the modules know they are supposed to be talking about the same object; the terminal state behaves as if the system knows it is finished; the message protocol behaves as if everyone knows that absence matters. None of them know any such thing, and the assumptions failed silently until an adversarial probe made them fail visibly. The repairs, when they came, were not injections of understanding; they were more competence, of the same humble kind (a comparison, a fraction, a third category in a taxonomy). The rationales for these mechanisms, meanwhile, float free of the system in Dennett's precise sense: the reason the veto works is that a mug's hum is part of what a mug is, and that reason is available to us and to no component of the system that enforces it. The degenerate objects of Section 6.4 mark the same boundary from the far side: a mug, to a competence, is wherever the bookkeeping becomes self-similar, and nothing in the system can notice that its own frame has a blind spot, because noticing is not among its competences. Comprehension, if it is ever to show up here, will presumably arrive the way everything else did: as one more mechanism with a rationale it cannot see.

8.2. Consciousness, ignored. A pattern ran through this work that we did not put there. The distinctions the engineering kept demanding are, one after another, the distinctions that discussions of consciousness usually claim as their subject matter. The difference between a sensor that receives nothing and a sensor that perceives an absence (Section 5) is the difference phenomenology has discussed since Sartre's *Pierre*, whose absence from the café is perceived, not inferred [9]. The fixation sequence that gives the retina its extent is a minimal attention mechanism. The consensus requirement is a unity-of-the-senses constraint, the binding problem restated as a termination criterion. The argument of Section 6.1, that recognition must be a continuously revised state rather than a terminated episode, is a claim about the specious present. In every case the distinction arrived as an engineering necessity, was implemented as arithmetic, and did its work with no phenomenal residue: nothing in Section 5 required there to be something it is like [8] to be the auditory module, and the veto functions identically whether or not the lights are on inside. Two readings of this pattern are available, and the data do not choose between them. On the deflationary reading, the ease with which these distinctions were mechanized is evidence that consciousness-talk was always a description of functional structure, never an explanation of it; we keep finding the functional shadow because the shadow is all there ever was. On the opposite reading, the fact that adversarial pressure forces exactly these distinctions, in this order, out of a system that began with none of them, suggests that the itinerary of consciousness is not arbitrary: the problems themselves demand this shape, and a research program that ignores consciousness will nevertheless be made to rebuild its outline, mechanism by mechanism, by nothing more mysterious than foils and ablations. We take no position here on which reading is correct, only note that "ignored" has functioned here as a methodology rather than an omission: at no point did any question of experience enter a design decision, and the resulting system nevertheless distinguishes measured absence from absent measurement, attends, binds, and refuses to conclude. Whether that is a debunking of the explananda or an independent rediscovery of them is, we think, the interesting question.

8.3. Knowledge as a property of a process: a first-author remark. One liberty available to an author outside the academy is to say where an idea actually came from. A year before this work I ran a project, informally called *Smart Guys*, that ended in what I took at the time to be a kind of machine-induced confusion; in retrospect it was the ordinary result of putting the wrong questions to the wrong parts of the world. What survived the episode was a single observation about Gettier's problem [10]. Since 1963 it has been known that the analysis of knowledge as justified true belief admits counterexamples, and the six decades since have been spent patching the definition. The counterexamples share a feature that the patches do not address: they are evaluated at an instant. Consider seeing a stranger across a room and concluding that a friend is present. At a glance the justification is impeccable, and suppose the belief is even true, because the friend is in fact standing behind you. Justified, true, and still not knowledge, since the justification never touched the fact that made it true. Sustained across a minute of walking beside the stranger, the same belief becomes a clinical matter, a delusion held against the evidence of the man at your side. Nothing in either man changed between the glance and the minute. What changed is that belief was allowed to run. The conclusion I drew is that the definition fails because it treats knowledge as a state satisfying conditions at a time, when it is the behavior of belief under continued sampling: belief is revised by the second, in parallel, across the sensory apparatus of a body, and knowledge is what remains standing. Time here plays the role of Dennett's universal acid [11]: poured over a belief, it eats through every justification that is not attached to the fact, and no container of definition holds it. What the acid leaves behind is knowledge, and nothing else survives the pour. Gettier cases, on this reading, are photographs of a process caught before it has run.

An hour before this section was written, that observation was used to untangle the communication between several sensory organs and the system they report to. The mechanics of Section 5 are the temporal reading

made operational. Once the system was required to wait, to sample the world over time rather than terminate at its first confident answer, the verdicts clarified on their own. Once the absence of evidence was read as positive evidence of a different world, one in which the predicted signal has no place, the false identifications on the probe set fell to zero without the system retreating to a safe alternative answer. The withheld verdict deserves particular attention as a feature of the design. In an embodied system it is the point at which the machine should turn to a human and ask what it is perceiving, receive an answer, investigate further, and retain the result. Contemporary large language models are trained toward the opposite disposition: premature confidence and reflexive agreement with the interlocutor. What one wants instead is a system that wants to learn, is able to learn, and is willing to stop, ask, and remember the answer. Stated in a few words, that is what was built here.

9. Limitations

The object set is two learned and three probe objects; none of the quantitative results should be assumed to generalize beyond it. The fixation policy uses the depth map to locate the object, a segmentation the sensors do not earn (the reference pipeline uses the semantic channel for the same purpose, a stronger oracle; the comparison of Section 2 is therefore internally consistent). CamSM has not been validated against the reference class on its native benchmark. The veto fraction (0.25) was set once and not swept. The consensus modifications are implemented at the termination decision because that is the available socket; we argue below they belong in the voting protocol.

10. Questions for the maintainers

1. Where should negative evidence live? The expected-signal test is attached to termination for lack of a better socket. Its natural home appears to be the voting protocol: a vote against a hypothesis, from a module whose modality that hypothesis makes predictions for.
2. Is there a planned representation for measured absence, that is, a prediction error carrying evidence weight when an expected feature is not found at a location, as distinct from the absence of an update?
3. Learning-module time appears to be the same question in another form: the present auditory graphs are invariant to tempo (the same object at any playback rate), and rests and gaps, which are measured absences over time, carry no evidence.

Reproducibility

```
# sensors and the three-module system
uv run python run.py experiment=triad_pretrain_2creatures
uv run python run.py experiment=triad_eval_2creatures
# (ablation matrix via dark/mute/numb overrides)

# representation comparison
uv run python run.py experiment=cam_pretrain_2creatures
uv run python run.py experiment=cam_eval_2creatures \
    env_interface=retina_eval_scaled      # also: rotated,
                                         # rotscale, tilted, tilted60

# Section 4 findings and Section 5 modifications
uv run python run.py experiment=triad_eval_2creatures \
    env_interface=retina_eval_foils      # finding 4.1
```

```
uv run python run.py experiment=triad_eval_consensus \  
    env_interface=retina_eval_foils          # 0 false identifications
```

Figures: `figures/triad.png`, `figures/eye_vs_eye.png`, `figures/where_refusal_lives.png`,
`figures/novelty_by_disagreement.png`, `figures/real_sonar.png`.

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